Materials and Design 31 (2010) 1599–1602

Contents lists available at ScienceDirect

Materials and Design

journal homepage: www.elsevier.com/locate/matdes

Short Communication

Effect of rare earth Y addition on the microstructure and mechanical properties of high entropy AlCoCrCuNiTi alloys

Zhaohua Hu, Yongzhong Zhan *, Guanghua Zhang, Jia She, Chunhui Li

Key Laboratory of Nonferrous Metal Materials and New Processing Technology, Ministry of Education, Guangxi University, Nanning, Guangxi 530004, PR China

ARTICLE INFO

Article history:
Received 8 July 2009
Accepted 9 September 2009
Available online 12 September 2009

ABSTRACT

A series of AlCoCrCuNiTiY$_x$ (x values in molar ratio, x = 0, 0.5, 0.8, 1.0) alloys have been prepared using vacuum arc melting. Classical high entropy diffraction peaks corresponding to a BCC crystal structure and some Cu, Cr peaks are observed for the AlCoCrCuNiTi alloy. However, with the incorporation of rare earth element Y, the BCC diffraction peaks disappeared and were replaced by new compounds like Cu_2Y and $AlNi_2Ti$. A typical cast dendrite structure with Cu-rich dendritic regions and Cr-rich rosette-like shape precipitations are found in the AlCoCrCuNiTi alloy. In the AlCoCrCuNiTiY$_x$ alloys, Y segregated preferentially to Cu and combined as bulky Cu_2Y compound. The maximum stress of the AlCoCrCuNiTi alloy is 1495 MPa, but reduces intensively after the incorporation of Y due to the formation of bulky Cu_2Y. For all the alloys, the compressive fracture mechanism is observed to be cleavage fracture.

1. Introduction

In the past the practical alloy systems were based mainly on one principle element as the matrix even though a substantial amount of other elements incorporated to enhance its properties or processability. It is anticipated that incorporation of multi-principle elements into alloys may result in the formation of many intermetallic compounds and complex microstructures, which would make alloys brittle and difficult in processing and analysis [1]. In recent years, Yeh et al. [2] have explored an entirely new class of alloy called high-entropy alloys (HEAs) with at least five principal elements and the concentrations between 5 and 35 at.%. HEAs are found to consist of simple crystal structure such as FCC and BCC, which results in promising mechanical properties (i.e. high hardness, large work hardening capacity and superior resistance to temper softening), wear and corrosion resistance [3–8].

Rare earth elements (REE) can greatly increase the properties of alloys such as tensile strength, ductility and anti-creep properties, so it is widely used in industry [9–11]. It would be of great interest to investigate the interaction of rare earth element in the HEAs. As HEAs is a type of newly developed alloy, however, its combination with REE has not been reported yet. In this paper, a series of rare earth added alloys AlCoCrCuNiTiY$_x$ (x values in molar ratio, x = 0, 0.5, 0.8, 1.0) are designed for the first time to investigate the effect of rare earth element Yttrium on the microstructural evolution and mechanical properties of the AlCoCrCuNiTi high-entropy alloy.

2. Experiments details

The ingots with nominal compositions of AlCoCrCuNiTiY$_x$ (x = 0, 0.5, 0.8, 1.0) were prepared from commercial-purity (all in 99.9 wt.%) Al, Co, Cr, Cu, Ni, Ti, and Y by electric arc melting under argon atmosphere on a water-cooled Cu hearth. The alloys were remelted at least five times in order to improve homogeneity. Finally, 15 mm diameter alloy button were prepared and the weight loss of all the four alloys is less than 1% after melting.

Structural investigations were carried out using X-ray diffraction (XRD) with Cu Kα radiation using a Rigaku D/Max-2500 V diffractometer. The Materials Data Inc. software Jade 5.0 and Powder Diffraction File (PDF release 2002) were used for phase identification. A Hitachi S-3400 N scanning electron microscope (SEM) equipped with an EDAX energy-dispersive X-ray (EDX) spectrometer was used for microstructural observation, phase composition identification and fractured surfaces observation. In order to evaluate the compressive properties, squares of $5 \times 5 \times 10$ mm were prepared from the 15 mm diameter as-cast alloy button, and a Zwick2.5 testing machine was used for compressive tests.

3. Results and discussion

3.1. Phase analysis

The XRD patterns of the as-cast alloy buttons are presented in Fig. 1. The AlCoCrCuNiTi alloy shows the presence of BCC solid-solution, Cu and Cr reflections, therefore it is a typical high-entropy alloy XRD patterns. However, with the incorporation of rare earth

* Corresponding author. Tel.: +86 771 3272311; fax: +86 771 3233530.
E-mail address: zyzmatres@yahoo.com.cn (Y. Zhan).

0261-3069/$ - see front matter © 2009 Elsevier Ltd. All rights reserved.
doi:10.1016/j.matdes.2009.09.016

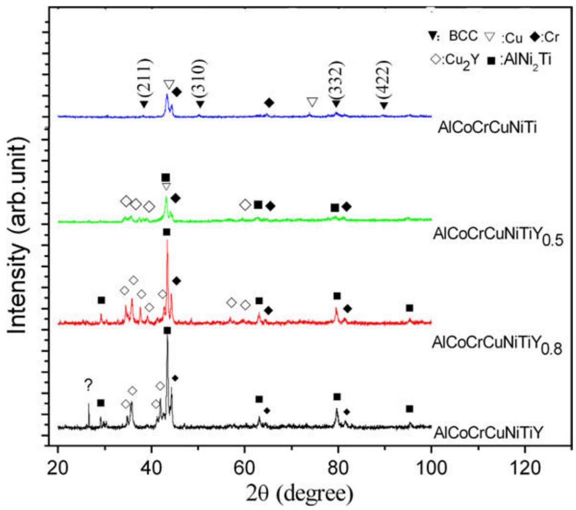

Fig. 1. XRD patterns of the AlCoCrCuNiTiY_x alloys.

element Y, the BCC solid-solution phase dissolved and some other crystal phases precipitated. In other words, the microstructure of AlCoCrCuNiTiY_x became rather complex. With the increase of Y content, the number of phases gradually increases. For AlCoCrCuNiTi$Y_{0.5}$ alloy, the BCC reflection disappeared and was replaced by the $AlNi_2Ti$ and Cu_2Y patterns. The microstructure of AlCoCrCuNiTi$Y_{0.8}$ alloy is similar to that of the AlCoCrCuNiTi$Y_{0.5}$ alloy, but its XRD intensity is much higher than the latter one. As for the equimolar AlCoCrCuNiTi$Y_{0.5}$ alloy, it has the highest relative intensity of all the four alloys. Besides the XRD patterns that exist in the AlCoCrCuNiTi$Y_{0.5}$ and AlCoCrCuNiTi$Y_{0.8}$ alloy, some unknown phases were found.

3.2. Microstructure

Fig. 2 shows the SEM images of the four as-cast alloys. The chemical compositions derived from EDX analysis are listed in Table 1.

The as-cast AlCoCrCuNiTi exhibits a typical cast dendrite structure with an average primary arm width of about 10 μm (as shown in Fig. 2a). The dendritic regions (marked as A) are mainly composed of Cu element with the atomic percentage of 74.77%. The interdendritic regions (marked as C) have nearly equimolar contents of Al, Co, Ni and Ti elements. Its structure is considered to be BCC according to the XRD pattern. In the interdendritic regions, a great amount of Cr-rich rosette-like shape precipitations were found (marked as B).

With the incorporation of Y element with x value of 0.5, the microstructure of AlCoCrCuNiTi$Y_{0.5}$ alloy changed obviously (as shown in Fig. 2b). Y element segregated preferentially to the Cu-rich dendritic regions (shown as Fig. 2b: B) and combined with Cu to form the Cu_2Y compound in a bulky shape (shown as Fig. 2b: A). The EDX data shown in Table 1, indicate that there are Al and Ni exist in A region, which is considered to be due to the high chemical activity of Y element. According to the XRD pattern of the alloy (shown in Fig. 1), the BCC phase were replaced by two intermetallic compounds i.e. Cu_2Y and $AlNi_2Ti$. The matrix is considered to be $AlNi_2Ti$. The Cr-rich rosette-like shape precipitation was not observed in the AlCoCrCuNiTi$Y_{0.5}$ alloy, it dissolved in the matrix (shown as Fig. 2b: C) as a solvent.

As the content of Y increased, in the AlCoCrCuNiTi$Y_{0.8}$ alloy (shown as Fig. 2c), the Cu-rich region reacted with Y and gradually

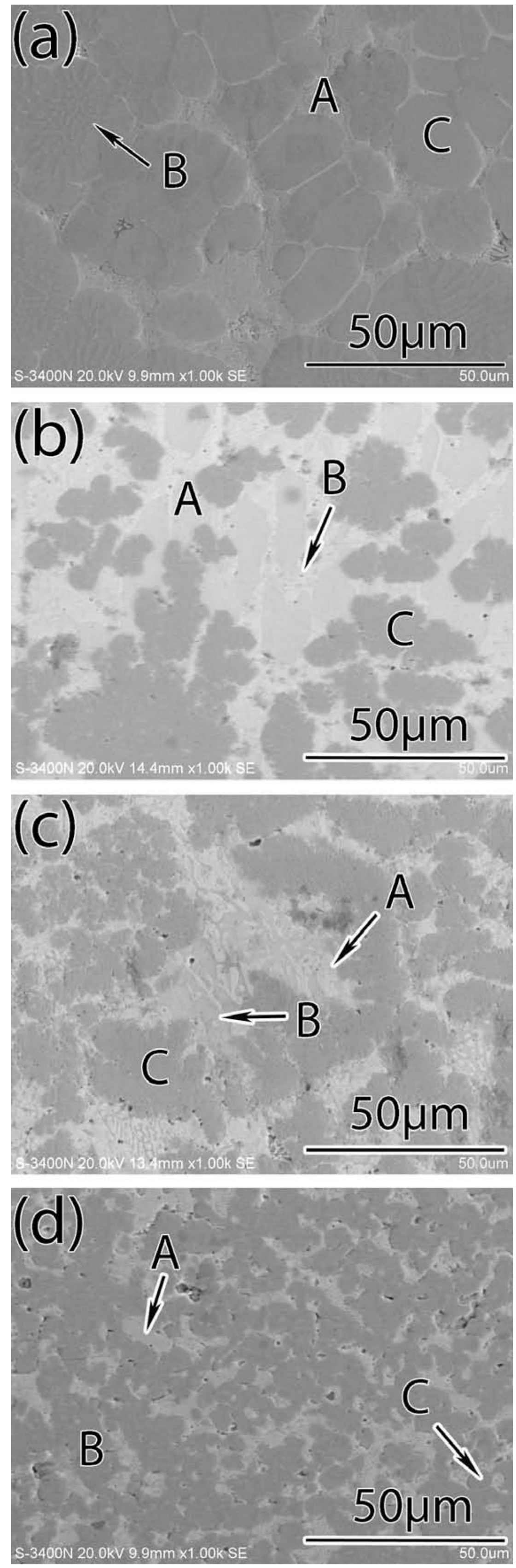

Fig. 2. SEM images of the as-cast AlCoCrCuNiTiY_x alloys, x values: (a) 0, (b) 0.5, (c) 0.8, (d) 1.

Table 1
Chemical composition of the $AlCoCrCuNiTiY_x$ alloys in atomic percentage.

Alloy	Al	Co	Cr	Cu	Ni	Ti	Y
AlCoCrCuNiTi							
A region	5.94	2.96	3.01	74.77	10.07	3.25	–
B region	4.37	7.89	77.37	1.43	4.05	4.89	–
C region	23.68	21.64	6.35	6.25	20.69	21.39	–
$AlCoCrCuNiTiY_{0.5}$							
A region	21.93	0.95	–	16.53	14.84	–	45.75
B region	4.63	1.11	–	34.73	11.44	–	48.09
C region	17.54	24.99	9.49	2.08	13.43	25.64	6.83
$AlCoCrCuNiTiY_{0.8}$							
A region	4.56	2.53	1.60	32.16	10.73	1.95	46.47
B region	18.98	1.02	0.58	17.03	14.25	0.35	47.79
C region	14.32	20.34	22.48	2.26	10.90	20.86	8.84
AlCoCrCuNiTiY							
A region	12.97	2.25	1.22	39.79	13.61	0.82	29.34
B region	21.88	22.48	3.40	3.03	19.17	23.44	6.60
C region	3.41	5.80	67.62	1.64	4.15	7.67	9.71

transformed to Cu_2Y (marked as B). The former bulky precipitated Cu_2Y was refined as plate shape (marked as A). As for the matrix, the Cr element dissolved in it completely and thus its content in the matrix increased up to 22.48%.

As for the equimolar AlCoCrCuNiTiY alloy (shown in Fig. 2d), the Cr element saturated in the matrix and precipitated again in a spherical shape (shown as Fig. 2d: C). According to the reaction formula $2Cu + Y = Cu_2Y$, the originally equimolar Y is excessive in the alloy, therefore the former dendritic Cu-rich region disappeared and was completely transformed into Cu_2Y. Moreover, the excessive Y dissolved in Cr and promoted its precipitation from the matrix. Meanwhile the precipitated Cr-rich zone retarded the growth of the bulky Cu_2Y, thus the Cu_2Y region (marked as A) was refined by the Y element apparently.

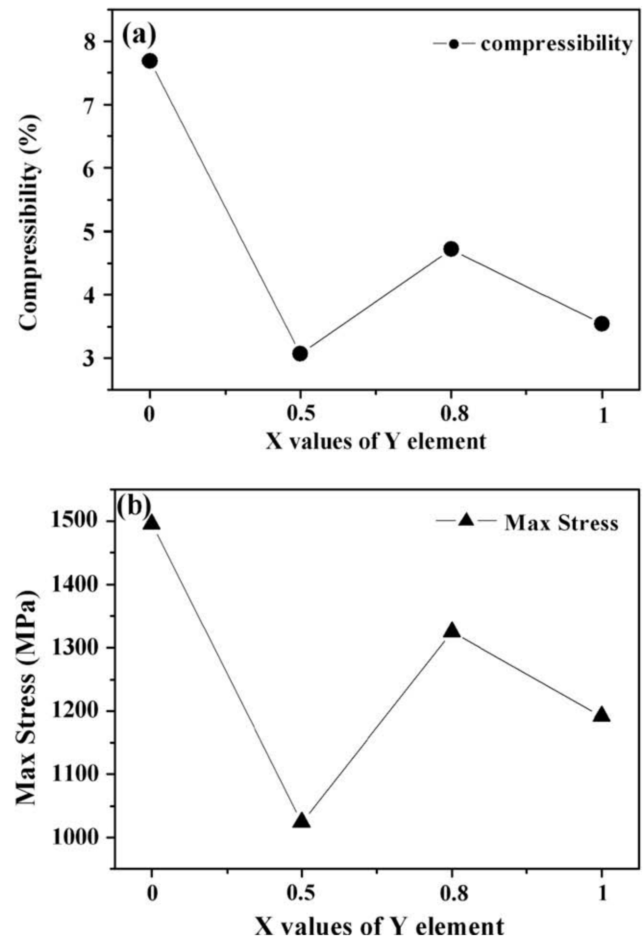

Fig. 3. Maximum stress and compressibility of the $AlCoCrCuNiTiY_x$ alloy.

3.3. Compressive properties

Fig. 3 shows the maximum stress and compressibility of $AlCoCrCuNiTiY_x$ alloy. The compressive strength and plastic strain limit of AlCoCrCuNiTi alloy are 1495 MPa and 7.68% respectively. It exhibits its high compressibility and high plastic deformability. With incorporation of Y, the compressive strength and plastic strain limit of the alloys reduced sharply. The $AlCoCrCuNiTiY_{0.5}$ exhibits the lowest compressibility of all the alloys. The Young's modulus (E), compressive strength and plastic strain limit for four alloys are listed in Table 2.

The fractured surfaces are shown in Fig. 4. For the AlCoCrCuNiTi alloy, the rough and quasi-cleavage facets show that its failure mode is typical intercrystalline fracture accompanied with cleavage fracture. Due to the low slip factor of BCC structure matrix and anchoring of dislocation by the Cr precipitation, it has the best compressive strength. Moreover, high plasticity dendritic Cu homogeneously dispersed in this alloy, thus it has a relatively high compressibility.

With the incorporation of rare earth element Y, the BCC solid-solution phase and the Cr phase dissolved. Besides, Y combined with Cu to form a brittle compound Cu_2Y, this directly leads to the brittleness of the $AlCoCrCuNiTiY_{0.5}$ alloy (as shown in Fig. 4b). From the fractured surface, clusters of dendrites along various directions can be clearly seen. The comminuted fractured surface indicates that the breakage firstly happened in the brittle bulky Cu_2Y compound, thus it has the lowest mechanical properties of the four alloys.

Table 2
Room temperature compressive properties of the $AlCoCrCuNiTiY_x$ alloy.

Alloy	E (Mp)	σ_{max} (MP)	ε_p (%)
AlCoCrCuNiTi	35,545	1495.0	7.68
$AlCoCrCuNiTiY_{0.5}$	35,815	1024.5	3.07
$AlCoCrCuNiTiY_{0.8}$	37,690	1324.5	4.73
AlCoCrCuNiTiY	36,855	1192.0	3.54

As the Y content increases, excessive rare earth element refined the crystalline and then strengthened the alloy, thus the compressive fracture mode of the $AlCoCrCuNiTiY_{0.8}$ alloy turned from transcrystalline fracture to intercrystalline. Consequently, the compressibility is improved as compared with the former one. The fractured surface is smooth and some minor quasi-cleavage facets are observed. The minor flaw size shows that the alloy has relative good ductility.

As for the equimolar AlCoCrCuNiTiY alloy, because of the fully combination of Y and Cu as the Cu_2Y compound, and the Cr precipitated again from the matrix as a bulky sphere shape, the compressibility of the alloy was reduced again. Furthermore, the intercrystalline fractured surface and bulky spallation shown in Fig. 4d indicate that the equimolar alloy also exhibits brittle cleavage fracture because of the existence of the multi-phases.

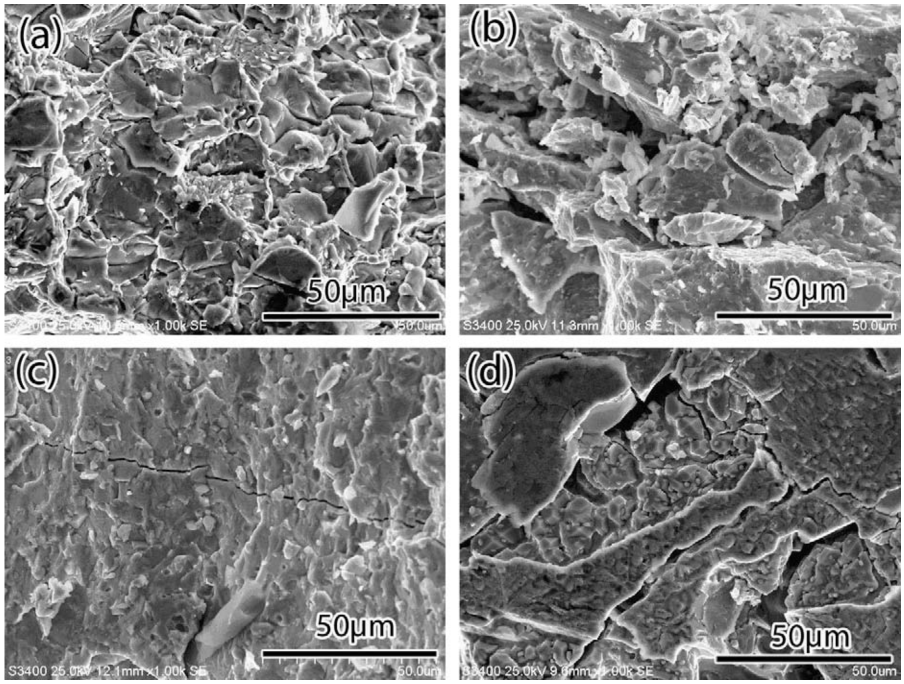

Fig. 4. SEM images of compressive fractured surfaces of the $AlCoCrCuNiTiY_x$ alloys: (a) AlCoCrCuNiTiY, (b) $AlCoCrCuNiTiY_{0.5}$, (c) $AlCoCrCuNiTiY_{0.8}$, (d) AlCoCrCuNiTiY.

4. Conclusions

In summary, the AlCoCrCuNiTi alloy has typical high entropy diffraction peaks corresponding to a BCC crystal structure and some Cu and Cr peaks. Besides, a typical cast dendrite structure with Cu-rich dendritic regions and Cr-rich rosette-like shape precipitations are observed. However, the BCC diffraction peaks were replaced by that of intermetallic compounds like Cu_2Y and $AlNi_2Ti$, after the incorporation of Y element. The maximum stress and compressibility of the AlCoCrCuNiTi alloy is 1495 MPa and 7.68% respectively, but reduced intensively after the incorporation of Y because of the brittle bulky Cu_2Y. All the alloys failed in brittle fracture, and the fracture mechanism was considered to be cleavage fracture.

Acknowledgements

The authors wish to express thanks to the financial support from the National Natural Science Foundation of China (50761003, 50601006), the Key Project of China Ministry of Education (207085) and the Opening Foundation of State Key Laboratory of Powder Metallurgy.

References

[1] Cantor B, Chang ITH, Knight P, Vincent AJB. Microstructural development in equiatomic multicomponent alloys. Mater Sci Eng 2004;375–377A:213–8.
[2] Yeh JW, Chen SK, Lin SJ, Gan JY, Chin TS, Shun TT, et al. Nanostructured high-entropy alloys with multiple principal elements: novel alloy design concepts and outcomes. Adv Eng Mater 2004;6:299–303.
[3] Huang PK, Yeh JW, Shun TT, Chen SK. Multi-principal-element alloys with improved oxidation and wear resistance for thermal spray coating. Adv Eng Mater 2004;6:74–8.
[4] Hsu CY, Yeh JW, Chen SK, Shun TT. Wear resistance and high-temperature compression strength of Fcc CuCoNiCrAl0.5Fe alloy with boron addition. Metall Mater Trans A Phys Metall Mater Sci 2004;35A:1465–9.
[5] Yeh JW, Chen SK, Gan JY, Lin SJ, Chin TS, Shun TT, et al. Formation of simple crystal structures in Cu–Co–Ni–Cr–Al–Fe–Ti–V alloys with multiprincipal metallic elements. Metall Mater Trans A Phys Metall Mater Sci 2004;35A:2533–6.
[6] Tong CJ, Chen SK, Yeh JW, Shun TT, Tsau CH, Lin SJ, et al. Microstructure characterization of AlxCoCrCuFeNi high-entropy alloy system with multiprincipal elements. Metall Mater Trans A Phys Metall Mater Sci 2005;36A:881–93.
[7] Tong CJ, Chen MR, Chen SK, Yeh JW, Shun TT, Tsau CH, et al. Mechanical performance of the AlxCoCrCuFeNi high-entropy alloy system with multiprincipal elements. Metall Mater Trans A Phys Metall Mater Sci 2005;36A:1263–71.
[8] Chen YY, Hong UT, Shih HC, Yeh JW, Duval T. Electrochemical kinetics of the high entropy alloys in aqueous environments – a comparison with type 304 stainless steel. Corros Sci 2005;47:2679–99.
[9] Stanford N, Barnett MR. The origin of "rare earth" texture development in extruded Mg-based alloys and its effect on tensile ductility. Mater Sci Eng 2008;496A:399–408.
[10] Dai QL, Luo CB, Xu XP, Wang YC. Effects of rare earth and sintering temperature on the transverse rupture strength of Fe-based diamond composites. J Mater Process Tech 2002;129:427–30.
[11] Xiao WM, Shi YW, Xu GC, Ren R, Guo F, Xia ZD, Lei YP. Effect of rare earth on mechanical creep–fatigue property of SnAgCu solder joint. J Alloys Compd. doi:10.1016/j.jallcom.2008.04.044.